

$$e) \quad x^3 y^2 (1 - x - y)$$

$$x^2 y^2 - x^4 y^2 - x^3 y^3$$

$$f(x) = 3x^2 y^2 - 4x^3 y^2 - 3x^2 y^3$$

$$f(y) = 2x^3 y - 2x^4 y - 3x^3 y^2$$

$$\begin{aligned} f(x) &= 3x^2 y^2 - 4x^3 y^2 - 3x^2 y^3 \\ &= x^2 y^2 (3 - 4x - 3y) \end{aligned}$$

$$x^2 y^2 (3 - 4x - 3y) = 0$$

$$4x + 3y = 3$$

$$x = \frac{1}{2}$$

$$\begin{aligned} f(y) &= 2x^3 y - 2x^4 y - 3x^3 y^2 \\ &= x^3 y (2 - 2x - 3y) \end{aligned}$$

$$x^3 y (2 - 2x - 3y) = 0$$

$$2x + 3y = 2$$

$$y = \frac{1}{3}$$